
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-Language Models (VLMs) offer a strong foundation for zero-shot goal-conditioned value estimation, but frozen pre-trained representations do not by themselves adapt to novel visual contexts or encode trajectory-level progress. We present VITA, a zero-shot value-function estimator that uses meta-learned test-time adaptation to combine semantic generalization with temporal context. At inference, VITA updates a lightweight adaptation module through a gradient step on a self-supervised reconstruction objective that is meta-trained to improve subsequent supervised progress prediction. Applied sequentially along a trajectory, these updates turn the module parameters into an implicit memory of preceding observations, enabling progress estimation without task-specific deployment demonstrations. We also introduce dissimilarity-based sub-trajectory sampling to reduce shortcut learning from redundant video frames. Across real-world robotic manipulation evaluations, VITA transfers from a single training environment to task, environment, and embodiment shifts, outperforming the state-of-the-art zero-shot autoregressive VLM baseline. The resulting value estimates further function as dense rewards for offline reinforcement learning: on Meta-World MT10, they yield multi-task policies whose performance exceeds policies trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) have become a central mechanism for cross-task and cross-domain generalization, acquiring transferable visual and linguistic structure from large-scale, weakly supervised web data (Radford et al., 2021; Alayrac et al., 2022). Generalist decision-making systems for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), however, continue to rely heavily on expert demonstrations (Ho & Ermon, 2016), which limits the scalability of task acquisition. World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) provide a complementary path by enabling agents to learn from simulated experience, with demonstrated relevance to robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). As such models increasingly generate realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), a more specific supervision problem comes to the foreground: *how can agents extract scalable learning signals from videos without human annotations or expert action labels?*

A direct approach to video-based decision making learns policies from expert visual trajectories by inferring latent actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Although these latent actions may ultimately be grounded in executable controls, learning that grounding remains a significant deployment bottleneck. A more reusable formulation is to learn a *goal-conditioned value function* that can be used zero-shot as a dense reward for reinforcement learning (RL) or as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this view, the value function converts visual trajectories and natural-language goals into task-progress supervision by estimating how far an observation has advanced toward completion (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Existing VLM-based value estimators expose a central architectural trade-off. Contrastive VLM methods (Radford et al., 2021) construct zero-shot rewards by comparing visual observations with

task descriptions in a shared multimodal representation space, thereby grounding progress estimates in semantic similarity (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Frame-level similarity, however, does not encode the trajectory-level temporal context required to distinguish visually similar states that occur at different task stages, such as folding versus unfolding a shirt. Autoregressive VLMs (Alayrac et al., 2022) can condition on a sequence of frames through the prompt, but their predictions can inherit monotonicity biases from chronologically ordered pre-training data (Ma et al., 2024). Consequently, both contrastive and autoregressive approaches remain constrained at zero-shot inference by pre-trained representations that are not explicitly adapted to the observed trajectory (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023). This constraint limits their ability to integrate semantic generalization with temporally grounded progress estimation (Pătrăucean et al., 2023). Other approaches increase adaptation capacity through large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demonstration-based reward specification (Sontakke et al., 2023), but these strategies require additional data, training, or task demonstrations.

We introduce VITA, a zero-shot value-function learning method that adapts a frozen vision–language representation at inference time in order to improve both semantic generalization and temporal reasoning, and that outperforms the state-of-the-art zero-shot autoregressive VLM approach (Ma et al., 2024) on real-world robotic manipulation tasks. Rather than requiring further large-scale pre-training or task-specific demonstrations, VITA adapts online to the semantic and temporal structure of each observed trajectory. At inference, a lightweight adaptation module is updated by a gradient step on a meta-learned self-supervised loss (Sun et al., 2024); during training, this update is optimized so that it reduces the supervised value-prediction loss (Finn et al., 2017). When applied sequentially over a test trajectory, the adaptation updates store trajectory history in the module parameters (Sun et al., 2024), yielding an implicit memory mechanism for temporally ordered progress estimation. We further introduce dissimilarity-based sampling to mitigate shortcut learning (Geirhos et al., 2020) by emphasizing semantically diverse trajectory segments during training. Our evaluation studies three uses central to goal-conditioned value functions: progress estimation under task, environment, and embodiment shifts; discrimination between expert and non-expert visual trajectories; and zero-shot reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation framework that turns frozen contrastive VLM representations into zero-shot goal-conditioned value estimators with trajectory-level temporal context, without task-specific deployment demonstrations or additional large-scale pre-training.
- We show that VITA transfers from a single training environment to out-of-distribution robotic manipulation tasks, environments, and embodiments, where it outperforms the state-of-the-art zero-shot value-function method (Ma et al., 2024).
- We demonstrate that VITA’s zero-shot value estimates can be used as dense rewards for offline RL, yielding Meta-World MT10 multi-task policies that surpass policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For visual trajectories, task-progress estimation can be cast as goal-conditioned value-function estimation by treating task completion as the underlying reward. We learn this value function from vision–language representations, using the predicted scalar as a normalized estimate of completion progress. Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, mapping an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to progress toward the goal. We set $V(o_t; g) = 0$ at task onset and $V(o_t; g) = 1$ at completion. Following common practice, expert-demonstration progress is aligned with temporal position, reflecting the assumption that expert trajectories make approximately monotonic progress toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, the supervised target is the normalized timestep index, $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T denotes trajectory length.

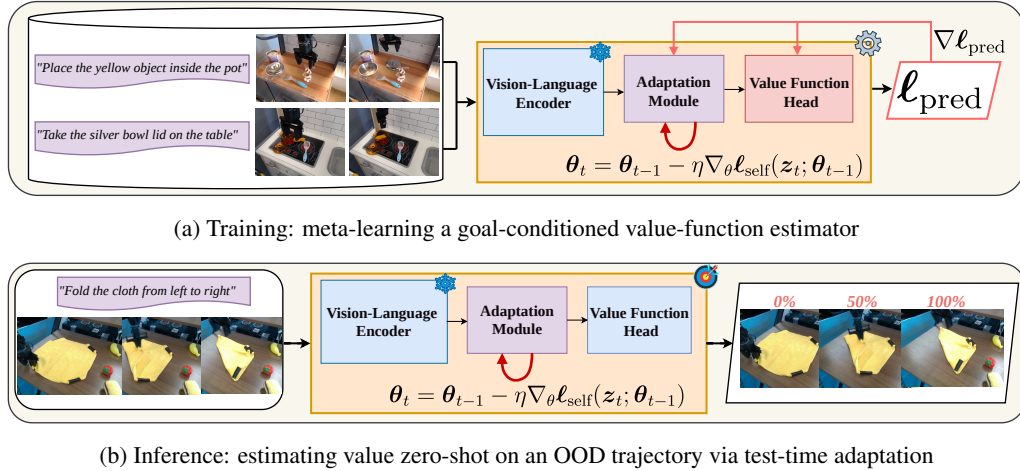


Figure 1: **Overview of VITA.** VITA meta-learns a goal-conditioned value estimator in which inference-time self-supervised updates provide trajectory-specific temporal context for zero-shot prediction on out-of-distribution trajectories.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference as a self-supervised learning problem (Goodfellow et al., 2016): model parameters are updated on each unlabeled test instance before prediction. Although TTT was first introduced for static image classification (Sun et al., 2020), subsequent work extends it to sequence modeling (Sun et al., 2024; Wang et al., 2023) by updating a parametric adaptation module f_{adapt} at every timestep with a gradient step on a self-supervised loss. Unlike conventional sequence models, which store temporal history in activation states such as recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT stores history directly in the parameters of f_{adapt} through sequential inference-time updates. These parameters therefore operate as an implicit memory that preserves temporal order while accumulating trajectory context. Sun et al. (2024) further meta-learn the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and optimized for the downstream prediction loss obtained after a test-time update, rather than for its reconstruction objective alone. We apply this mechanism to goal-conditioned value-function learning, where sequential updates provide temporal context and the self-supervised task is meta-learned specifically to improve value estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

The VITA estimator decomposes goal-conditioned value prediction into three functionally distinct modules: (1) a frozen multimodal encoder that embeds trajectory observations and task descriptions; (2) a test-time adaptation module whose parameters are modified by a meta-learned self-supervised objective; and (3) a regression head that converts the adapted representation into a scalar estimate of goal progress.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We instantiate the multimodal encoder with a frozen contrastive vision-language model, CLIP (Radford et al., 2021), which supplies separate image and text representations for visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate the visual embedding and the goal embedding at each timestep t , producing the multimodal sequence $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders. The contrastive pre-training objective of CLIP aligns paired

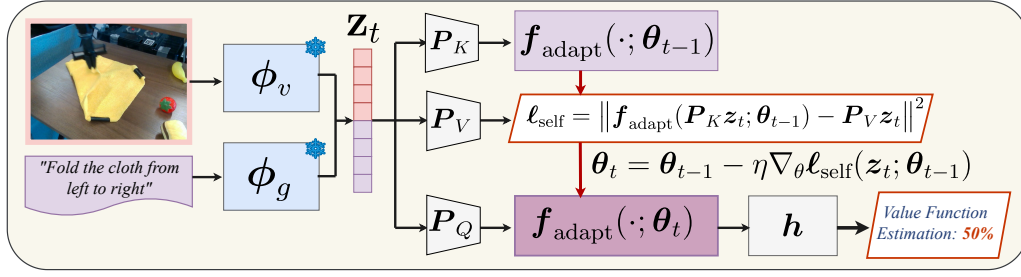


Figure 2: **Test-time Adaptation.** During inference, each timestep t updates the adaptation module f_{adapt} with a gradient step on the meta-learned self-supervised loss ℓ_{self} , thereby encoding temporal history in the module parameters.

image–text inputs in a shared representation space (Radford et al., 2021); VITA uses this semantic alignment as the input representation and relies on test-time adaptation to add trajectory-specific temporal context.

3.1.2 TEST-TIME ADAPTATION

To condition the frozen multimodal representation on the semantic and temporal context of the observed trajectory, VITA applies an adaptation module f_{adapt} using the test-time training procedure introduced in Section 2.2. During inference, f_{adapt} is updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} . The loss is a reconstruction objective parameterized by learnable linear projections P_K and P_V ; these projections are optimized during training so that the resulting inference-time update improves downstream value estimation rather than merely solving a generic reconstruction task. Specifically, $P_K \in \mathbb{R}^{d' \times 2d}$ produces the input view and $P_V \in \mathbb{R}^{d' \times 2d}$ defines the target representation, yielding the self-supervised loss:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the adaptation-module parameters immediately before processing timestep t . The online update is

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Because updates are applied sequentially rather than independently, the parameters of f_{adapt} carry information from earlier observations forward to later predictions. Section 4.6.2 empirically compares this parameter-memory mechanism with trajectory-level, memoryless, and explicit-window alternatives.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After adaptation at timestep t , a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps z_t into the adaptation space $\mathbb{R}^{d'}$. The projected representation is processed by f_{adapt} , and a regression head h converts the adapted feature into a scalar goal-conditioned value prediction:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), following value-estimator architectures commonly used in deep reinforcement learning (Espeholt et al., 2018). Training uses the normalized expert-demonstration progress labels y_t described in Section 2.1; the supervised prediction loss ℓ_{pred} is the mean squared error between $V(z_t; g)$ and y_t . At inference, h remains fixed, while f_{adapt} is updated through the self-supervised objective. This design separates a stable value readout from a trajectory-adaptive representation, enabling zero-shot prediction without task-specific demonstrations.

3.2 TRAINING PROCEDURE

The training objective is designed to match the adaptation operation used at inference. We optimize the value estimator with gradient-based meta-learning, learning the self-supervised loss ℓ_{self} so that the parameter update it induces reduces the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Meta-training batches are constructed from dissimilarity-selected sub-trajectories, thereby exposing the model to visually and semantically varied temporal segments rather than to highly redundant adjacent frames.

3.2.1 META-LEARNING

During training, f_{adapt} is updated at each timestep using ℓ_{self} , while the outer objective differentiates through these updates with respect to the supervised prediction loss ℓ_{pred} , following gradient-based meta-learning (Finn et al., 2017). This procedure jointly optimizes the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total loss combines ℓ_{pred} and ℓ_{self} , with the self-supervised term weighted by λ . At inference, the learned initialization θ_0 serves as the starting point for online adaptation, so the model is explicitly trained to make its self-supervised test-time updates predictive of value rather than merely reconstructive.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories frequently contain extended sequences of near-duplicate consecutive frames, a form of temporal redundancy also documented in video action recognition (Wang et al., 2018). In value-function learning, such redundancy can encourage shortcut behavior (Geirhos et al., 2020; Lee et al., 2021b): the estimator may associate progress with recurrent late-stage appearances rather than with semantic evidence of task advancement. We counter this tendency with dissimilarity-based sampling, which constructs mini-batches from visually diverse sub-trajectories within each trajectory. The selected batches increase intra-batch variation and serve as an importance-sampling heuristic that places greater training weight on semantically informative temporal segments.

Formally, for a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we construct a candidate set $\mathcal{W}$ by extracting fixed-length sub-trajectories with a sliding window of size w_{tr} and stride s . The desired subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i denotes a sub-trajectory of length w_{tr} . Exact optimization of Eq. 4 is combinatorial, since it requires enumerating all $\binom{|\mathcal{W}|}{k}$ subsets. We therefore use an efficient scoring heuristic that assigns each window $w \in \mathcal{W}$ its total dissimilarity to all candidate windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Selecting the k windows with the largest diversity scores yields a diverse subset in polynomial time under this heuristic. Appendix G shows that the associated computational overhead is negligible relative to model training. The ablation in Section 4.6.1 isolates the empirical role of dissimilarity-based sampling: compared with full-trajectory sampling, it improves discrimination between expert and non-expert visual trajectories.

4 EXPERIMENTS

We evaluate VITA through three complementary tests of whether a goal-conditioned value function remains useful beyond its training distribution: (i) progress estimation for real-world robotic manipulation under task, environment, and embodiment shifts; (ii) discrimination between expert and non-expert visual trajectories within the same robotic domain; and (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark.

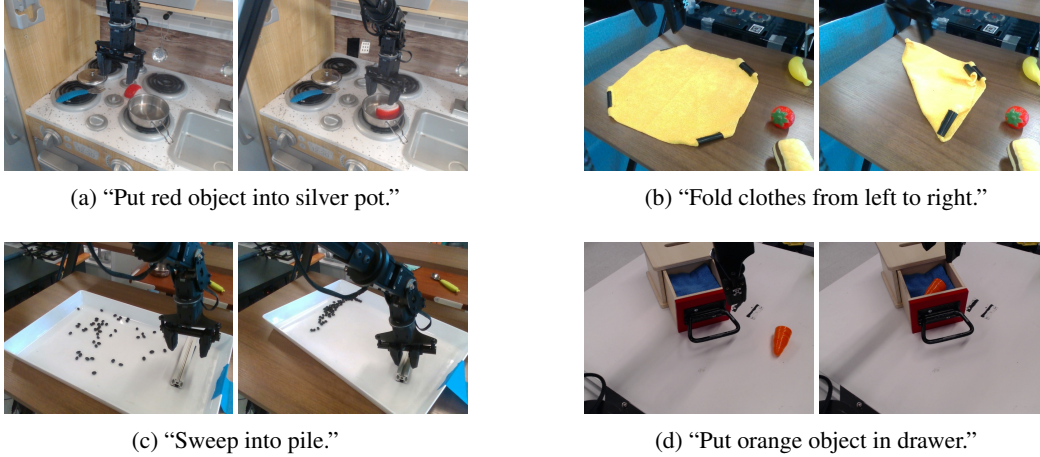


Figure 3: Representative visual trajectories and task descriptions from the distribution-shift evaluation. (a) In-distribution pick-and-place. (b, c) Environment-shifted manipulation tasks. (d) Combined embodiment and environment shift.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions from BridgeData V2 (Walke et al., 2023), a dataset spanning manipulation tasks, environments, and robot embodiments. The training split is intentionally narrow relative to the evaluation suite: it contains 2,986 expert demonstrations of pick-and-place tasks, excludes folding, sweeping, and stacking, and is collected with a single WidowX 250 embodiment across 4 ToyKitchen configurations. We reserve 287 additional expert demonstrations as the in-distribution test set, denoted τ_{k_pnp} ; Appendix A shows representative training examples. OpenCLIP ViT-B/32 is used as the frozen backbone for video-frame and task-description encoding. The objective combines ℓ_{pred} and ℓ_{self} , with $\lambda_{self} = 0.5$. During training, dissimilarity-based sampling uses window size $w_{tr} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, adaptation uses one gradient step ($t_{ep} = 1$) with learning rate $\eta = 0.1$. Appendix D reports further implementation details and hyperparameter sweeps.

4.2 BASELINES

We compare VITA with contrastive and autoregressive VLM-based value estimators, as well as supervised temporal baselines trained on the same frozen CLIP representation. VLM-CL produces zero-shot values from the cosine similarity between CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, while CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same frozen representation sequence. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value-function method based on autoregressive VLMs. We evaluate GVL-0S in the zero-shot setting, where the prompt contains only the test trajectory and task description, and GVL-1S in the one-shot in-context setting, where the prompt additionally includes a full shuffled training-distribution trajectory with progress labels. Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with the proposed prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Appendix D provides additional implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

To assess generalization within real-world robotic manipulation, we curate BridgeData V2 subsets that vary task, environment, and robot embodiment relative to the training distribution. Environment shifts change scene layout or background: $1m_pnp$ evaluates pick-and-place in front of a laundry machine;

Table 1: Value Order Correlation (VOC) for progress estimation under task, environment, and embodiment shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

td_fold, ft_fold, and rd_fold evaluate cloth folding on different surfaces; and ms_sweep evaluates a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, whose morphology and camera perspective differ from the training embodiment (WidowX 250). dt_tk_pnp and dt_tk_stack retain the in-distribution ToyKitchen environment with the new embodiment, whereas dt_ft_stack and dt_rd_pnp combine embodiment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except ms_sweep, which contains 100 due to its size limitation; Appendix B gives the full dataset description. We report *Value Order Correlation* (VOC) (Ma et al., 2024), which measures how well predicted progress aligns with frame chronology in a visual trajectory. Let $p_1, \dots, p_T$ be the predicted progress values for a T -frame trajectory and $r_t = t$ the temporal index. VOC is the Spearman rank correlation ρ_s between predicted values and temporal indices, and therefore evaluates temporal ordering rather than absolute value calibration.

Table 1 shows that the methods with explicit temporal mechanisms, VITA and CLIP-GRU, consistently outperform the autoregressive VLM baselines GVL-0S and GVL-1S on VOC. This comparison highlights a central empirical pattern in our evaluation: preserving and modeling temporal order is more reliable for progress estimation than prompting a pre-trained autoregressive VLM with shuffled or in-context trajectories. Among the temporal methods, VITA achieves the higher VOC on 6 of 10 evaluation datasets. Both VITA and CLIP-GRU also substantially exceed CLIP-based baselines without temporal modeling (CLIP-FT, VLM-CL, VLM-RM), indicating that frame-level semantic similarity is insufficient for these progress-estimation settings. CLIP-FT is competitive with GVL on the in-distribution task but does not transfer to the out-of-distribution settings, while VLM-CL and VLM-RM perform poorly. For GVL-0S and GVL-1S, in-context examples sometimes improve performance, sometimes provide no benefit, and sometimes reduce it. The GVL variants perform well on folding datasets (td_fold, ft_fold, rd_fold) but do not reach comparable performance on stacking (dt_tk_stack, dt_ft_stack) or pick-and-place (dt_tk_pnp, lm_pnp), suggesting that the autoregressive VLM used by GVL may be biased toward folding-like manipulations. VITA, by contrast, maintains strong performance across task types and shift categories, although the margins over CLIP-GRU vary by dataset.

The long-horizon ms_sweep task is difficult for nearly all methods, yet VITA obtains the highest VOC score and exceeds CLIP-GRU, supporting the value of encoding history through sequential parameter updates rather than only through recurrent hidden states in this setting. Under embodiment shift, VITA also remains robust. On dt_tk_pnp, which changes the robot embodiment while preserving the environment and task family, both VITA and CLIP-GRU exceed their in-distribution performance, showing that learned temporal context can transfer across embodiments in this setting. By contrast, GVL-1S performs poorly on dt_tk_pnp and dt_tk_stack, indicating that one-shot in-context prompting does not reliably handle embodiment shifts in our evaluation.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond distributional generalization, a useful progress estimator should distinguish expert behavior from trajectories that share the same environment and task family but do not exhibit expert-like progress. We therefore compare predictions on expert and scripted non-expert trajectories from the in-distribution BridgeData V2 setting. The scripted trajectories are generated in ToyKitchen with a heavily randomized controller (Walke et al., 2023), using the same embodiment and environmental configuration as the expert demonstrations. They cover pick-and-place tasks over object categories that can overlap with training, including general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task, embodiment, and environment, their behavior departs from the smooth, monotonic progress typical of expert demonstrations; Appendix C provides examples. Because the scripted data are produced by a randomized controller, they do not encode calibrated levels of suboptimality. Nevertheless, an effective value estimator should assign lower VOC to these suboptimal trajectories than to expert trajectories in the same setting. We quantify this property with *BinVOC*, defined as $1 [\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when expert trajectories obtain higher VOC than suboptimal trajectories and 0 otherwise.

We average BINVOC over the scripted datasets and compute expert-trajectory performance (VOC_{exp}) on the in-distribution `tk_pnp` test set, whose pick-and-place tasks match the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination, assigning higher VOC to expert trajectories for every scripted task. VITA also exceeds CLIP-GRU on this binary criterion, suggesting that the implicit memory induced by sequential test-time updates is less vulnerable to this scripted shortcut setting than the recurrent hidden-state baseline. The CLIP-similarity baselines (VLM-CL, VLM-RM) underperform, consistent with their lack of temporal modeling. CLIP-FT is more reliable than those zero-shot similarity baselines but fails on one task, indicating sensitivity to object-distribution variation.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We next test whether the learned value estimator can serve as a zero-shot dense reward for offline RL. The evaluation uses Meta-World MT10 (Yu et al., 2020), which contains ten robotic manipulation tasks including pick-and-place, door opening, and pushing. VITA is trained only on real-world robotic data and is then applied zero-shot in this simulated benchmark. For each MT10 task, we generate 20 expert visual demonstrations with Meta-World expert policies and use the goal-conditioned value estimator to assign a dense reward to each visual observation. Multi-task policies are trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021). We repeat expert-data generation and policy training over 10 random seeds; after training, each policy is evaluated on 20 rollouts per task, and returns are averaged for each (task, seed) pair. Following Agarwal et al. (2021), we report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals. Appendix E details the training and evaluation protocol.

Table 2 reports MT10 performance under this protocol. VITA obtains the highest IQM return, 0.815, and outperforms every CLIP-based reward baseline under the same IQL setup. We do not report a direct GVL comparison at this scale because GVL depends on Gemini-1.5, a proprietary autoregressive VLM whose inference cost is prohibitive for RL reward generation; the open-source alternatives we tested did not provide reliable progress estimates. As a strong task-specific reference, the fuzzy-logic dense reward supplied by Meta-World (META-WL) achieves 0.779. VITA surpasses this simulator reward in the reported IQM comparison, demonstrating that a value estimator trained on real-world visual trajectories can provide an effective reward signal for simulated offline RL in this benchmark.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the effect of dissimilarity-based sampling by comparing sub-trajectory selection strategies on the ability of VITA to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$; in our setting, this exposes the model to global temporal regularities that can become

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Expert-versus-scripted discrimination, measured by average BinVOC across 5 in-distribution scripted datasets; higher values indicate more reliable ordering of expert trajectories above scripted non-expert trajectories. (b) Offline RL performance on Meta-World MT10 under the same IQL protocol, measured by interquartile mean (IQM) with 95% stratified bootstrap CIs.

shortcuts. Random sampling increases coverage by selecting sub-trajectories uniformly, but it does not explicitly favor semantic diversity. Dissimilarity-based sampling instead selects sub-trajectories with high pairwise dissimilarity and yields the strongest discriminative performance, outperforming both full-trajectory and random sampling on expert–non-expert separation. Appendix F.1 provides the full quantitative comparison.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We next analyze how different test-time adaptation strategies for incorporating temporal context affect value-function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} . This procedure encodes trajectory history directly into the adaptation parameters, preserving temporal order while accumulating context. We compare VITA with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update computed over the full trajectory. This produces a batched update that averages the loss across all frames, thereby discarding temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step. By adapting only to the current visual observation, this approach prevents any temporal information from being carried forward; (iii) *explicit memory (TTT-EX)*, which also resets the adaptation module at every step but updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). Across all datasets, VITA outperforms trajectory-level, memoryless, and explicit-memory methods, showing that incremental parameter updates provide the most effective temporal context among the tested alternatives. We provide the full analysis in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been applied as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), using frame-level similarity scores but lacking temporal modeling. Some works fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to generate dense rewards, requiring supervision that limits their zero-shot applicability. RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning, but yields sparse rewards and still requires demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on scale for generalization. ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, combined with a few in-domain demonstrations to improve generalization. In contrast, VITA adapts a pre-trained contrastive vision–language encoder at inference, capturing both temporal and semantic context through test-time adaptation, without requiring large-scale multimodal pre-training or additional domain-specific demonstrations for generalization. VITA could be applied to any pre-trained vision–language representation that encodes a universal value function to enhance

its generalization and temporal reasoning via meta-learned test-time adaptation. Autoregressive VLMs (Alayrac et al., 2022) leverage in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) mitigates monotonic bias from ordered pre-training data by shuffling frames at inference, but this discards temporal order, which is essential for temporal reasoning and for distinguishing visually similar yet temporally distinct states. Our approach instead preserves chronological order while capturing temporal context through test-time adaptation, avoiding such shortcuts.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers a parameter-efficient route for modifying VLM behavior without full fine-tuning. Prompt-based methods optimize task-conditioned parameters at inference, although the mechanisms by which these prompts alter model behavior are often difficult to interpret (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) suggested that CLIP representations could support test-time adaptation, based on the observation that CLIP encodes human-interpretable concepts discovered via mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also been used as a reward signal, with parameters optimized at test time via reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) provides a more direct mechanism: an adaptation module is updated at each timestep through a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training enables online adaptation at inference without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

The experiments support test-time adaptation as a practical mechanism for zero-shot value estimation in robotic manipulation under task, environment, and embodiment shifts. Beyond monotonic progress estimation on expert demonstrations, VITA distinguishes expert from scripted non-expert visual trajectories and supplies dense rewards that improve offline RL performance under the reported MT10 protocol. Its central mechanism is sequential parameter memory: by updating the adaptation module at each timestep, the model accumulates trajectory history in its parameters and uses that history to condition later value predictions. The comparison with CLIP-based baselines, CLIP-GRU, and autoregressive VLM baselines indicates that this implicit memory provides a useful alternative to both frame-level similarity and recurrent hidden-state modeling for zero-shot value-function estimation. At the same time, the gains over the strongest non-autoregressive baseline are sometimes incremental, so the evidence is best interpreted as showing consistent and parameter-efficient improvement across the studied settings rather than universal dominance. In our implementation, the inference-time overhead remains small because only the lightweight adaptation module is updated at test time.

6.2 LIMITATIONS AND FUTURE WORK

Several limitations delimit the scope of these findings. The supervised progress labels are derived from expert demonstrations and therefore inherit an approximately monotonic relation between time and task completion; trajectories with pauses, reversals, retries, or extended failures may require additional modeling beyond the present formulation. Although the adaptation module is lightweight, updating a value estimator at every timestep introduces a deployment consideration for real-time closed-loop systems, especially when execution safety or latency constraints are stringent. The real-robot evaluations also rely on the datasets and splits described in this paper, which makes clear release and reproducibility details important for subsequent comparison. Future work should study alternative test-time adaptation objectives for vision-language models, evaluate the method in closed-loop control and more complex RL environments, and analyze theoretically how diversity-based sampling affects shortcut learning. These directions extend the core contribution of VITA: a goal-conditioned value estimator that meta-learns test-time adaptation to improve zero-shot temporal reasoning from visual trajectories.

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A TRAINING DATASET

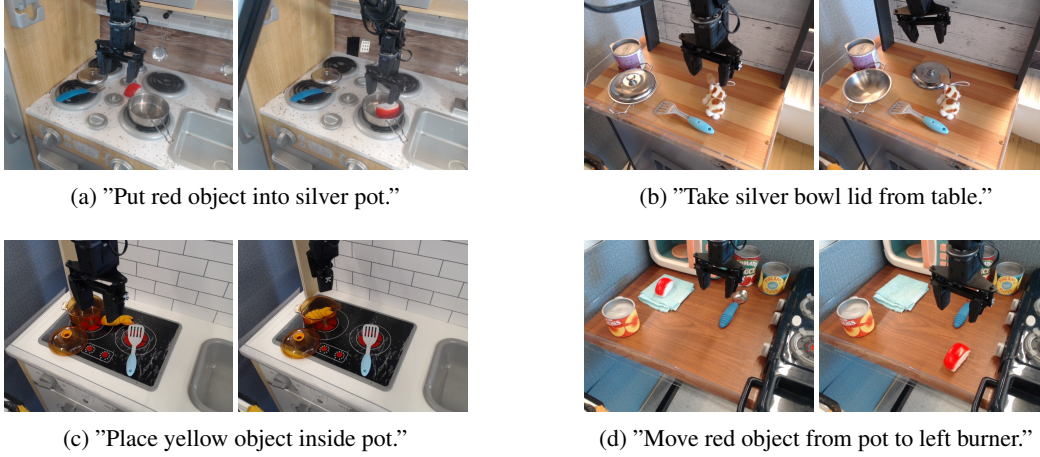


Figure 4: Each subfigure presents the first and final frames of an expert training demonstration together with its natural-language task description. The training demonstrations span four ToyKitchen environments.

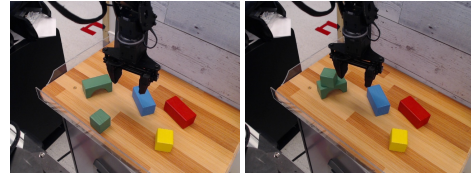
B EVALUATION DATASETS

Table 3: Evaluation datasets organized by task type, visual environment, and robot embodiment.

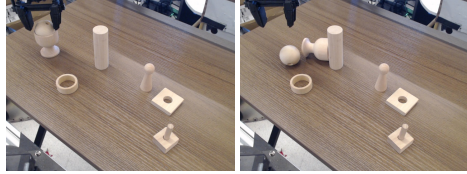
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

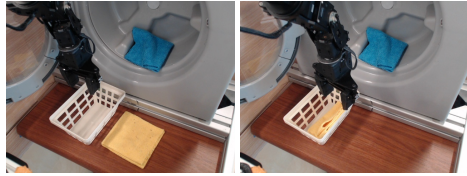


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure shows the first and final frames of an evaluation trajectory under embodiment shift, together with the associated natural-language task description. The top row isolates an embodiment shift within ToyKitchen by replacing the training robot with DeepThought, whereas the bottom row combines the embodiment shift with new environments.



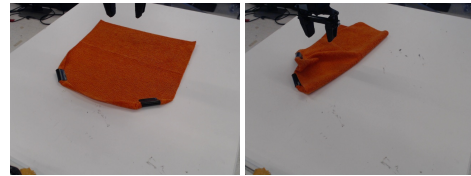
(a) lm_pnp: "Place blue cloth inside washer."



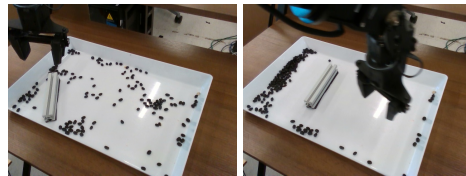
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure shows the start and end frames from an evaluation trajectory under environment shift, along with its natural language task description. All tasks are performed using the same robot embodiment across visually and structurally different environments.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure shows the start and end frames from a scripted evaluation trajectory collected in the ToyKitchen environment using the WidowX 250 robot. All tasks involve pick-and-place behavior with varying object categories. We use generic task descriptions due to the lack of publicly available annotations.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone. Each visual observation and task description is encoded using the CLIP vision and text encoders to produce their representation, respectively. We opted for a joint (concatenation-based) representation over an element-wise product based on improved validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and a projection dimension $d' = 64$. The model is trained using the AdamW optimizer with a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , and a cosine learning rate schedule with 10% warmup. We use a batch size of 32 and pad all trajectories to a maximum length of 120 frames (matching the longest sequence in the training set). The weighting coefficient λ_{self} of the self-supervised loss in the total training objective is set to 0.5, selected based on validation performance. We train for 5 epochs, as validation VOC typically plateaus early, with extended training providing no further improvement. For dissimilarity-based sampling, we set the stride to $s = 1$, the sub-trajectory length to $w_{\text{tr}} = 8$, and the number of selected windows to $k = 8$, unless otherwise specified. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time, we adapt only the temporal adaptation module f_{adapt} , using the same projection dimension ($d' = 64$) as in training. We perform a single gradient step ($t_{\text{ep}} = 1$) with

Table 4: Effect of test-time learning rate η and number of adaptation steps t_{ep} on VITA performance on the validation set measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

a learning rate $\eta = 0.1$. This configuration was selected from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$, based on validation performance reported in Table 4. The results show that both (i) a single update with a larger learning rate ($\eta = 0.1, t_{\text{ep}} = 1$) and (ii) multiple updates with a smaller learning rate ($\eta = 0.01, t_{\text{ep}} = 10$) achieve comparable validation performance under test-time training. We select the former because it requires fewer adaptation steps while achieving slightly better results.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rates η of 0.1, selected via hyperparameter tuning as described previously. CLIP-FT follows VITA’s architecture, but excludes the adaptation module and self-supervised loss. In particular, CLIP-FT uses a frozen CLIP encoder, followed by a linear projection and a two-layer MLP to predict task progress. The model is trained with standard supervised regression, without meta-learning or test-time adaptation. To increase expressivity, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt that describes the environment. CLIP-GRU follows VITA’s architecture (frozen CLIP encoder and regression head h), but replaces the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) module that explicitly encodes temporal history through recurrent hidden states. The GRU module has a single layer with hidden size 64, and its final hidden state is passed to the regression head h . For both GVL-0S and GVL-1S, we use the latest version of Gemini 1.5 Pro, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), whereas the original GVL implementation used an earlier release, `gemini-1.5-pro`. We also evaluated GVL using the open-source autoregressive VLM `Qwen-VL 2.5` (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations followed the prompt template introduced in the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with the following settings, kept fixed across all tasks. Expectile regression is set to $\tau = 0.7$, with advantage weighting temperature 3.0 and clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies are trained for 100,000 gradient steps, with evaluation every 10,000 steps on 20 rollouts per task (horizon 150). Implementation follows `d3rlpy`, with configuration: `expectile=0.7, weight.temp=3.0, max.weight=100.0, actor.lr=1e-4, critic.lr=3e-4, batch.size=256`.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM computes the mean of the middle 50% of scores, which reduces sensitivity to outliers compared to the mean or median. For statistical robustness, we estimate

95% confidence intervals using stratified bootstrap with 10,000 resamples, stratified by task. This procedure is applied consistently across all methods in Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC results across scripted datasets under different training-time sampling strategies for test-time adaptation. A checkmark indicates successful discrimination between expert and suboptimal trajectories. Sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We empirically validate the effectiveness of the proposed dissimilarity-based sampling by comparing the impact of different sub-trajectory sampling strategies. In particular, we evaluate how different sub-trajectory sampling strategies during training impact the ability of our model to discriminate expert from suboptimal trajectories. In Table 5, we compute Disc@VOC for 5 scripted datasets, capturing whether a given model assigns a higher average VOC to expert trajectories (`tk_pnp`) than to each evaluation dataset. We compare three strategies: (1) full-trajectory sampling applies adaptation over all overlapping sub-trajectories of length w_{tr} using stride s , following prior work (Sun et al., 2024); (2) random sampling uniformly samples k sub-trajectories of length w_{tr} per video; and (3) our proposed dissimilarity-based sampling constructs a candidate set of sub-trajectories of length w_{tr} using stride s , and selects k sub-trajectories that maximize pairwise dissimilarity in the representation space.

Our results show that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ yields perfect discriminative performance. Sampling fewer windows ($k = 4$) of size $w_{tr} = 8$ leads to the same performance, while reducing sub-trajectory length to $w_{tr} = 4$ reduces effectiveness. The full-trajectory baseline performs the worst, confirming that adapting over all consecutive windows can lead to overfitting to global temporal cues. Random sampling outperforms full-trajectory sampling but remains less reliable than dissimilarity-based selection. These findings highlight the importance of both sub-trajectory diversity and semantic locality in enabling discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We evaluate four adaptation strategies: (1) TTT-TR, which performs a single gradient update computed over the full trajectory (offline). TTT-TR performs a batched update that averages the loss across all frames, thereby discarding temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates using only the current visual observation (memoryless). TTT-RS prevents any temporal information from being carried forward; (3) TTT-EX, which resets the adaptation module at every step and updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). TTT-EX averages gradients over this window, discarding the temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory). VITA forms an implicit memory through the sequential accumulation of temporal history in the adapter parameters. We performed the same test-time hyperparameter sweeps for all variants, as described in Appendix D. We found that VITA, TTT-RS, and TTT-TR achieve optimal performance with a learning rate of 0.1 and a single adaptation step, whereas TTT-EX performs best with a learning rate of 1.0 and one adaptation step.

Table 6: Validation VOC scores for progress estimation under distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

The results in Table 6 show that TTT-TR, TTT-RS, and TTT-EX yield comparable performance. This indicates that performing batched updates—whether over a short explicit window (TTT-EX) or over the full trajectory (TTT-TR)—performs similarly to the memoryless baseline (TTT-RS). In contrast, VITA consistently outperforms all three variants, underscoring the importance of sequential, per-timestep updates for effective value function estimation in real-world robotic manipulation tasks. In implicit memory (VITA), updating over a local window of frames would cause the most recent observations in that window to be incorporated twice—explicitly through the window and implicitly through the accumulated parameters. In addition, updating less frequently (non-sequentially) would cause information loss because intermediate frames are omitted. Therefore, the implicit-memory test-time training paradigm requires sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets as described in Eq. 4, we adopt a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others (i.e., the row sum of the distance matrix). Let the number of candidate windows be $N_c = T - w_{tr} + 1$, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, which requires $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, max length 120). With a batch size of 32, feature dimension $d = 1024$ (CLIP-based multimodal representation), and window length $w_{tr} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This cost is small relative to a single forward-backward pass through the model and is incurred to promote the selection of diverse training windows.